FACTOR

Dominican Digital ID

Security Boundary Map

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Dominican Digital ID — Security Boundary Map

Detailed structural evidence of the cards examined, compared boundary by boundary. All seventeen boundaries (0–16). This detailed structural comparison describes only those cards; the findings on the RSA-1024 development signer and on the CSCA trust gap were confirmed separately on production cédulas (see the assurance and RSA-1024 reports) and are not generalized beyond the cards examined to the full issued population.

#	Boundary	Current status	Requirement to pass / validate it	Security consequence
0	Physical transport	**Passed**	Physical access to the reader; PC/SC activation and T=1 transport	Card activation only
1	Public discovery	**Passed**	Ordinary access to the card	Read of MF, EF.DIR, EF.ATR/INFO and EF.CardAccess
2	Contactless secure channel	**Passed**	Printed CAN and PACE parameters	Authenticated encrypted session; no persistent change
3	Ordinary LDS identity data	**Passed**	Active CAN/PACE secure session	Read of the permitted identity data
4	Passive authenticity verification	**Partial**	JCE CSCA, revocation information and validation policy	Integrity verified (valid SOD signature; 7/7 readable DGs reconcile with SHA-256); issuer trust unresolved
5	Chip authentication	**Passed**	CAN/PACE, CA-v1 AES-256/P-384 suite and key ref. 0x30	Possession proven live; the chip retains its non-exportable private key
6	DG3 fingerprint access	**Not yet passed**	Authorized CVC chain CVCA → DV → terminal (CAR `DOCVCA00001`), terminal private key and DG3 rights	High-sensitivity biometric access; SELECT returns 6982 without EAC-TA
7	PKCS#15 general information	**Passed**	CAN/PACE	SSCD `DomRep SSCDv1`; full decoding: the ODF declares the 9 directories; two non-extractable private keys (signature + authentication), six authentication objects (PIN/PUK), one 3DES administration key descriptor and minidriver data are provisioned; only the certificate directory (CDF) is empty
8	General user key operation	**Partial**	User PIN (ID 05/ref. 83) — held and verified — plus an operational key DF	The user PIN verifies (9000), but key 0x42 resides in the PKCS#15 DF (5015), which is in the initialization lifecycle state (0x03): INTERNAL AUTHENTICATE returns 6982. Key use is blocked by the DF state, not by the PIN
9	Dedicated SSCD signature	**Not yet passed**	Signature PIN value (ref. 92), not delivered to the holder, and a certificate	The signature DF (0DF5) is *operational* (0x05) and the non-exportable signature key is present; blocked only by the undelivered signature PIN and the missing certificate
10	Citizen certificate enrollment	**Partial**	JCE electronic signature enrollment: identity verification, certificate issuance and delivery of the signature PIN	The signature PIN/PUK are provisioned on the card but their values *were not delivered* to the holder (the issuance record carried only the user PIN and the SO PIN); no certificate — the digital signature is not yet enrolled
11	Signature recovery	**Not yet passed**	Signature PUK (ID 23/ref. 93) and authorized recovery protocol	Persistent change of the credential state
12	PKCS#15 administration	**Passed**	SO PIN (ID 07/ref. 84)	SO PIN held and verified: administrative access to the user/5015 domain (manages the user PIN). It does not reach the signature domain (0DF5); the signature PIN is governed by the signature PUK, not the SO PIN
13	External administrative authentication	**Not yet passed**	192-bit 3DES administrative secret (ref. 0x21) and challenge–response protocol	Key object observed in the SKDF (authId 07); its use requires the secret, which is not held
14	DomRepEID proprietary children	**Passed**	CAN/PACE; select the DomID AID	Read-only sweep (SFI 1–30): the DF maps to the MF and re-exposes only public files (EF.ATR/INFO, EF.CardAccess, EF.CardSecurity, EF.DIR); no personal data or new secrets
15	GlobalPlatform / JCOP management	**Not yet passed**	Issuer Security Domain keys, SCP configuration and management privilege	Control of applets, privileges and lifecycle
16	Manufacturing / personalization	**Not yet passed**	Vendor profile, issuer keys, HSM/SAM authority and production systems	Creation or replacement of security objects; destructive issuer authority

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Passed	Partial	Not yet passed	Incomplete	Unknown
Read in full using only the physical card and the information printed on it (or, in 12, with a PIN delivered to the holder).	The test progressed without issuer credentials, but does not confirm the whole layer.	Not yet passed in this test; requires an issuer secret or a certificate that is not held.	The function appears not to be configured on the cards observed.	Not enough accessible evidence to determine its status.

Independent production cédula evidence. Boundaries 2, 3 and 5 were exercised live on a production cédula: CAN/PACE was established, the identity data groups were read and Chip Authentication was completed (possession of the non-exportable private key proven). Boundary 4: the CMS signature of the EF.SOD is valid and the seven readable data groups (1, 2, 7, 11, 12, 13, 14) reconcile with the SOD under SHA-256; issuer trust remains unresolved because the JCE cédula CSCA is not retrievable (the distribution host ca.jce.gob.do answers NXDOMAIN — reconfirmed with a live query; the same host serves the DS's CRL distribution point, so revocation checking is equally unreachable). A full-record decoding of boundary 7 (which corrects an earlier truncated read) shows that the card is fully personalized: the ODF declares all nine directories; the PrKDF has two non-exportable private keys (signature, ref. 0x41, signature PIN; authentication, ref. 0x42, user PIN); the AODF has six authentication objects and a live status query confirms that every PIN/PUK is set (SO, user, signature PIN and signature PUK, with full retry counters); and the 3DES administration key and the minidriver data are present. The only missing element is the on-card X.509 certificate (empty CDF), also empty on other cards examined — probably kept off the card or pending, not a per-card deficiency. Every boundary reachable without an issuer secret or a card-verifiable certificate (0–5, 7, 14) has been read in full; the rest require credentials held by the JCE.

Credential and lifecycle evidence. The two codes on the issuance record were correlated with the chip: the "PIN (initial)" is the user PIN (ref. 83) and the "PUK" is the SO PIN (ref. 84) — both held and verified (9000); the SO PIN opens boundary 12 (administration of the user/5015 domain) but has no rights in the signature domain. The signature PIN (ref. 92) and the signature PUK (ref. 93) are provisioned (a status query returns a retry counter, and wrong attempts decrement it — a stored value exists), but their values were never delivered to the holder, and there is no certificate. The lifecycle is split: the signature DF 0DF5 (signature key, signature PIN and PUK) is operational (0x05), while the PKCS#15/authentication DF 5015 (authentication key, user PIN and SO PIN) is in initialization (0x03). Consequently, the signature key is not blocked by any lifecycle state — only by the undelivered value of the signature PIN and the missing certificate (boundary 9) — while the authentication key fails with 6982 because its DF is still in initialization (boundary 8). Overall: the electronic signature is not yet enrolled; its PIN slots are pre-provisioned, but the credentials and the certificate must come from a JCE signature enrollment step.

Live retry counters (non-mutating query, never consumed): SO PIN 5/5, user PIN 3/3, signature PIN 3/3, signature PUK 5/5. **Source:** the project's detailed internal boundary-map log (not included in this published repository) and the tools/active_adaptive/ toolset.